

USER GUIDE

HIAS

Hall Invigilator Allocation System

St. Xavier's Catholic College of Engineering (Autonomous), Nagercoil

Version **1.1.0** · Windows desktop application · Works fully offline

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1. What HIAS does

HIAS assigns invigilators (staff) to examination halls for every exam session, and rotates those assignments fairly. It also seats the students of each session in the same halls, two to a bench, and prints the seating plan and attendance sheets. It remembers who has already been in which hall, so nobody is sent to the same hall again until they have had a turn in every other hall of the rotation.

Set up once Departments, halls and staff › **Create a batch** Exam dates and FN/AN sessions › **Generate** Pick staff and halls; HIAS rotates them › **Confirm** Rules are checked automatically › **Print** Export the allocation and reports as PDF

- **Fully offline.** HIAS needs no internet connection and no server. Everything is stored on the computer it is installed on.
- **No login.** The app opens straight to the dashboard; there are no accounts or passwords inside HIAS. Anyone who can use this computer's Windows account can use HIAS, so the protection is the Windows account and physical access to the computer: keep HIAS on a computer (and Windows account) that only authorised people use.
- **Automatic rule checks.** Before an allocation is confirmed, HIAS checks that no hall or person is double-booked, nobody repeats a hall too early, and inactive staff or halls are not used.

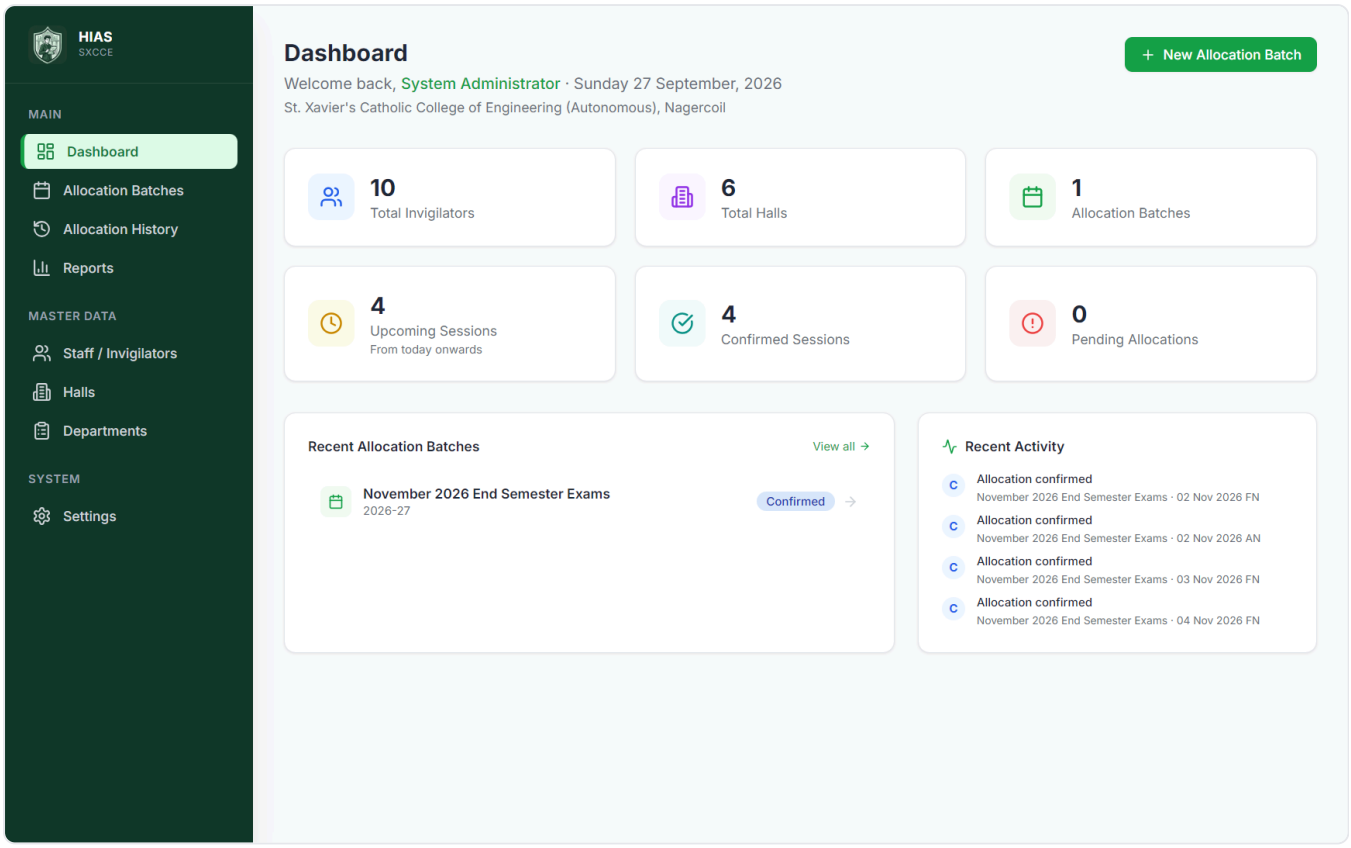
2. Installing HIAS

1. Copy `HIAS-Setup-1.1.0.exe` (about 76 MB) to the computer and double-click it.
2. If Windows shows "*Windows protected your PC*", click **More info** and then **Run anyway**. This appears because the installer is not code-signed; it is safe.
3. When Windows asks "*Do you want to allow this app to make changes to your device?*", click **Yes**. HIAS installs for everyone who uses the computer, which needs administrator permission once.
4. Choose the installation folder (the default is fine) and click **Install**.
5. Leave **Run HIAS** ticked and click **Finish**. Next time, open HIAS from the **HIAS** shortcut on the desktop or in the Start menu.

Installing a newer version over an older one upgrades it and keeps all data. To remove HIAS, use Windows **Settings → Apps → Installed apps → HIAS - Hall Invigilator Allocation System → Uninstall**.

> **Where is my data?** All data is kept in one file on this computer: `%APPDATA%\eias\eias.db` . Updating or reinstalling HIAS keeps this file. Uninstalling the app does not delete it either. See section 10 for backups.

3. A tour of the screen



The dashboard. The sidebar on the left is always available. (All names in this guide are sample data.)

Sidebar item	What it is for
Dashboard	Totals at a glance: invigilators, halls, batches, upcoming and confirmed sessions, and sessions still pending allocation. Click a card to jump to it.
Allocation Batches	Create an exam batch (for example "November 2026 End Semester Exams") and open it to allocate each session.
Allocation History	Search every confirmed allocation by batch, session, staff member or hall.
Reports	Staff-wise, date-wise, timetable and audit reports, each exportable as PDF.
Staff / Students / Halls / Departments	The master data HIAS allocates and seats from.
Settings	College name, default session timings, backup and restore, and rotation restart.

A **Back** link at the top of each page returns to the previous screen. The button at the top of the sidebar shrinks it to icons (hover over an icon to see its name); HIAS remembers your choice. In a narrow window, for example when HIAS fills half the screen, the sidebar shows icons automatically. The bottom of the sidebar reads **Offline · Local database**: everything you see is stored on this computer.

4. First-time setup

Do these steps once, in this order. You can come back and change anything later.

4.1 College profile and session timings

Open **Settings** → **College Profile**. The **College Full Name** is printed at the top of every PDF; the **Short Name** appears under the HIAS logo in the sidebar. Click **Save**.

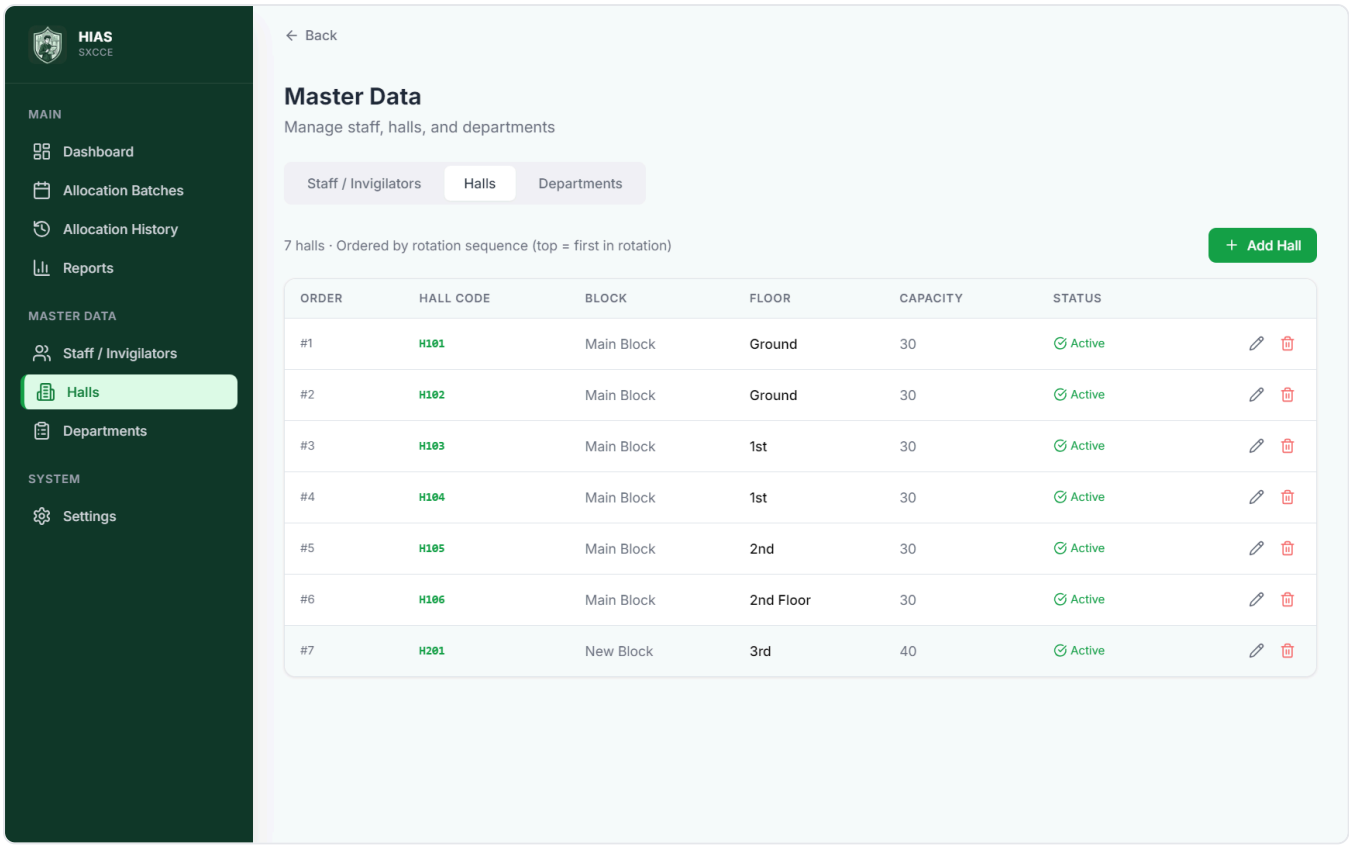
Under **Settings** → **Session Timings**, set the usual reporting, start and end times for the Forenoon (FN) and Afternoon (AN) sessions. New batches start with these times; you can still change them per session.

4.2 Departments

Open **Departments** and add each department with a short **Code** (for example `CSE`) and a full **Name**. Reports use the short code. The code is saved exactly as you type it, but capital and small letters count as the same code: once `CSE` exists, `cse` or `Cse` is refused as a duplicate. The bin icon removes a department that no staff member belongs to, after asking you to confirm.

4.3 Halls

Open **Halls** → **Add Hall**. Only the **Hall Code** (for example `H101`) is required. **Block** and **Floor** (Ground, 1st, 2nd…) say where the hall is; they are printed next to the hall code on the allocation sheet so invigilators know where to go. Under **Bench layout (student seating)**, enter the number of **Columns** of benches (left to right) and, as **Rows**, the most benches any column has (front to back). A grid appears, drawn as seen from the board: tap a bench that is not really there (a pillar, the door) to remove it, or tap a column or row heading to remove or restore the whole line. Benches are numbered column by column, front to back, and two students sit on each, so the capacity is worked out for you. A hall with no bench layout can still have an invigilator but cannot seat students. Untick **Active** for a hall that is temporarily unavailable; inactive halls are never allocated.



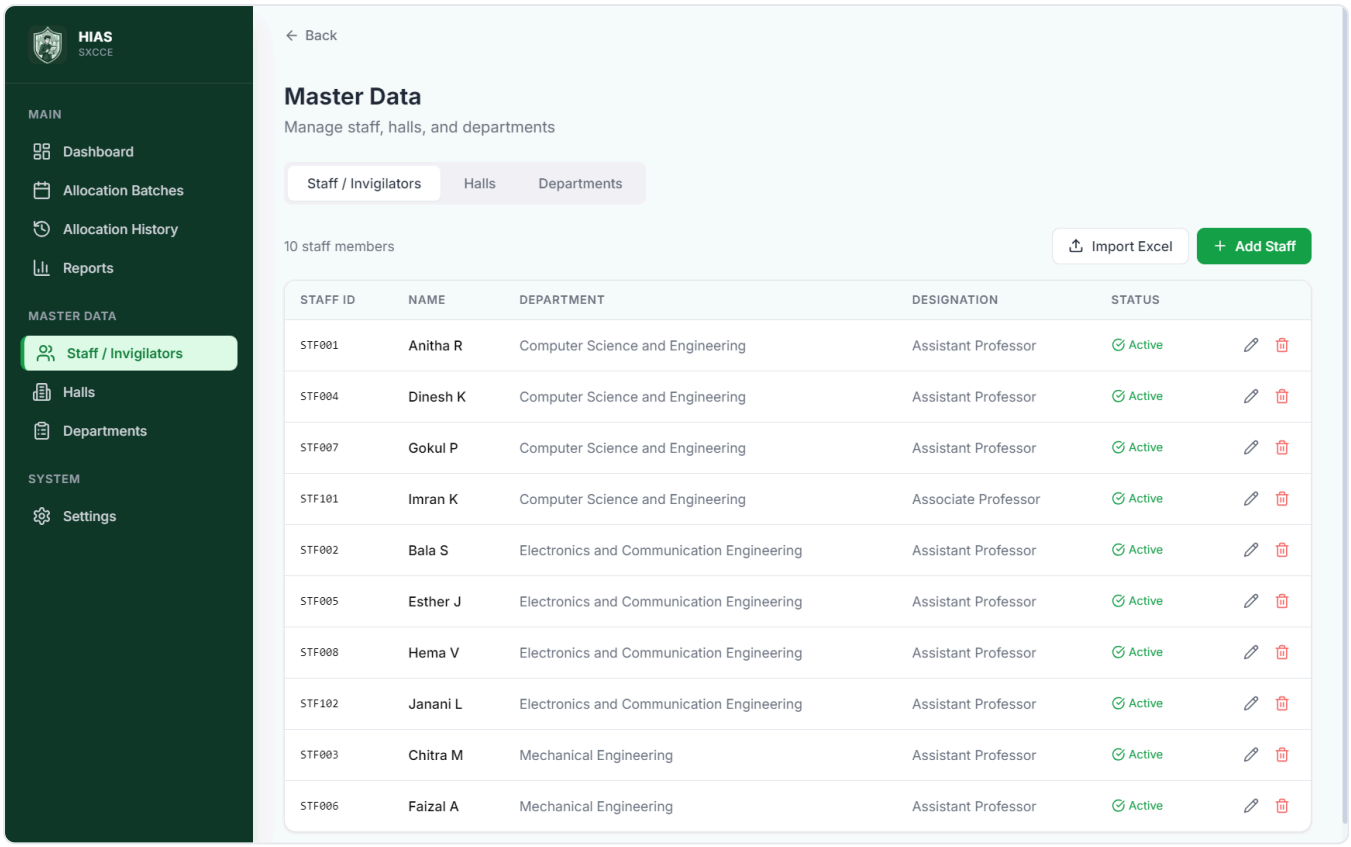
Halls. The pencil edits a hall; the bin removes a hall that has never been allocated, after asking you to confirm (for a used hall, untick Active instead).

4.4 Staff (invigilators)

Add staff one at a time with **Add Staff**, or many at once with **Import Excel**. The file `HIAS-Staff-Import-Template.xlsx`, supplied with HIAS, has the right headings ready. For an import, the first row of the sheet must contain these column headings:

Column	Required?	Notes
Staff ID	Yes	Must be unique. Rows whose Staff ID already exists are skipped.
Name	Yes	Full name of the staff member.
Department	Yes	The department's code or full name; it must already exist in Departments.
Designation, Email	No	Optional details.

After the import, HIAS shows how many staff were added and lists each skipped row with the reason (for example "*Row 5: Department CIVIL does not match any existing department*"), so you can correct the sheet and import it again.



Staff / Invigilators, with Import Excel and Add Staff at the top right.

> **Deactivate or delete?** The bin icon first **deactivates** a staff member: they stay in past records but are no longer allocated. An inactive staff member with no allocation history can then be **deleted permanently**; HIAS asks you to type `DELETE` to confirm.

4.5 Students

Open **Students** in the sidebar. Add students one at a time with **Add Student**, or many at once with **Import Excel**; click **Template** first for a sheet with the right headings:

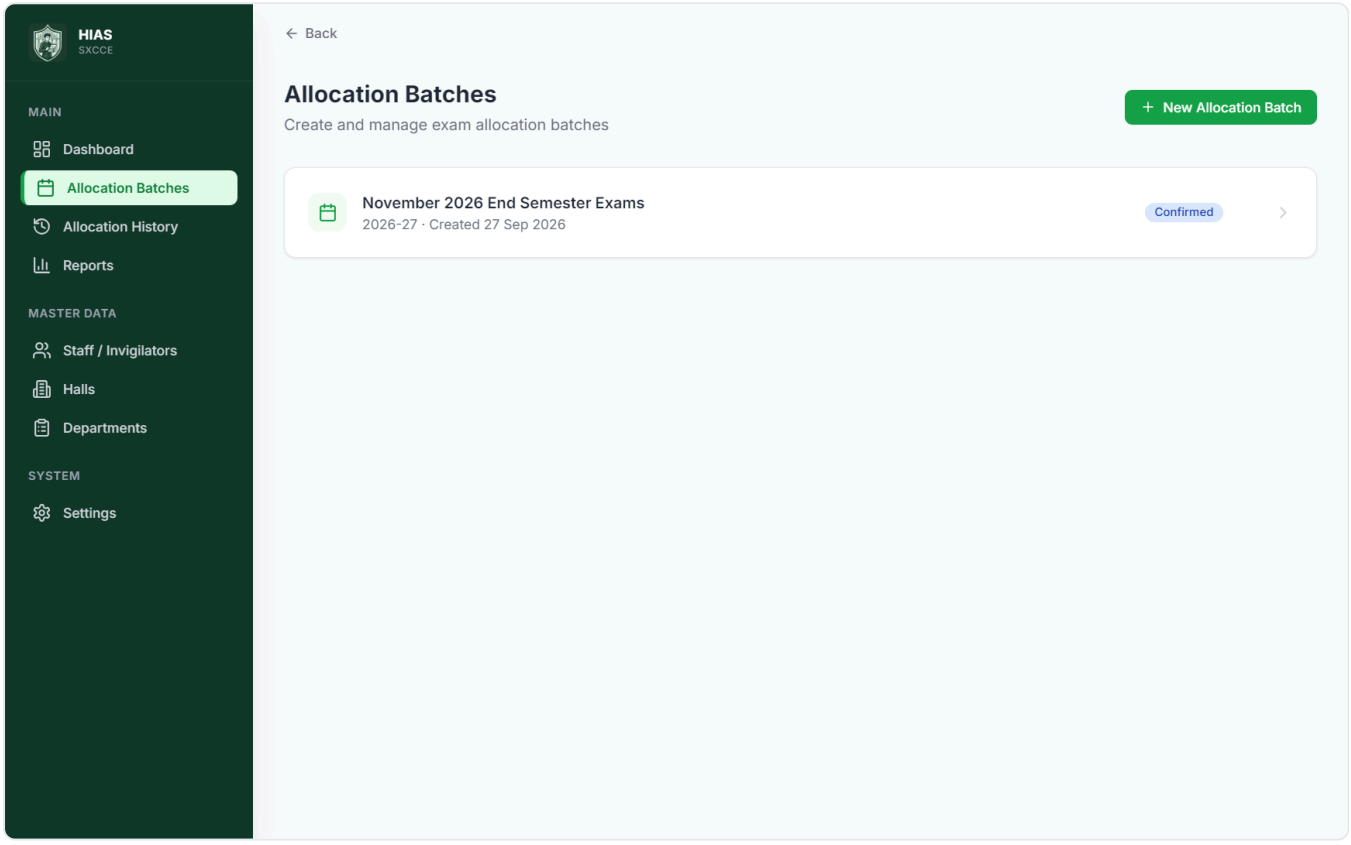
Column	Notes
Roll No	Unique, for example `102301`. Seat columns are filled in roll-number order.
Register No	Unique university register number.
Name	Full name of the student.
Gender	`Male` or `Female` (`M` / `F` and `Boy` / `Girl` are also accepted).
Department	The department's code or full name; it must already exist in Departments.
Batch	The years of study, for example `2023-2027`. HIAS works out the year (Final, Third, Second…) from the batch and the academic year of the exam batch.

As with staff, rows that cannot be used are listed with the reason after the import. In seating, a **class** means one department in one batch, for example *CSE – Third Year*.

5. Creating an allocation batch

A **batch** is one examination series, such as an end-semester or internal assessment exam. Open **Allocation Batches** → **New Allocation Batch**. The wizard has three steps:

1. **Batch details.** Enter the batch name, the academic year (for example `2026-27`) and the number of exam days.
2. **Pick the dates** on the calendar. Sundays are highlighted and cannot be chosen, so you do not pick them by mistake.
3. **Sessions and timings.** For each date, choose Forenoon (FN), Afternoon (AN) or both. Timings come from Settings; expand a date to change its times.



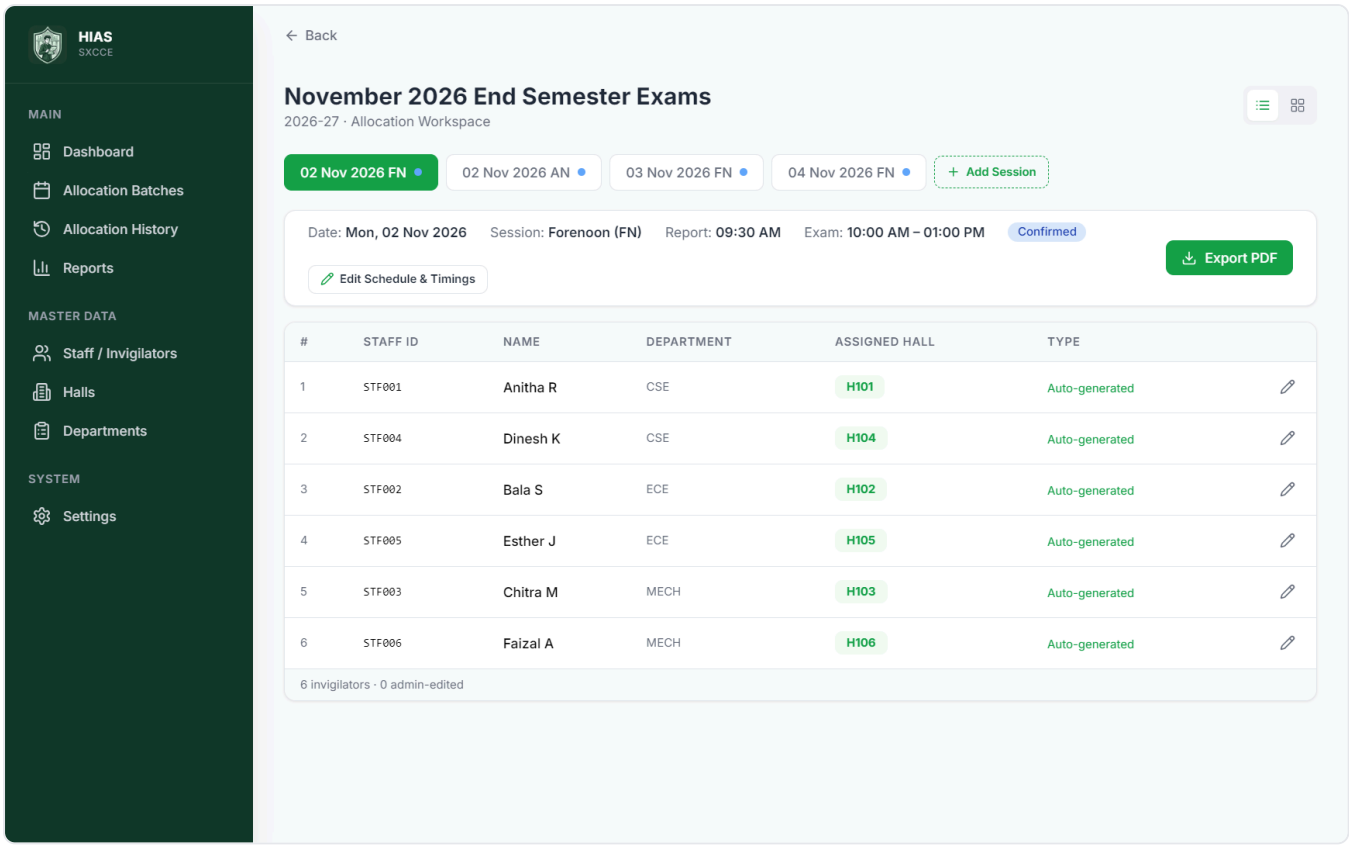
Allocation Batches. A batch shows **Draft** until every session in it is confirmed, then **Confirmed**.

To delete a whole batch, open **Settings** → **Delete Batch**, click **Delete** beside the batch, type the batch name and enter your own name and Staff ID. They must match an active person in the Staff list (a made-up name or a wrong Staff ID is refused), so only registered staff can delete a batch. HIAS keeps a record of every deleted batch (its name, when it was deleted and who deleted it), listed under **Deleted batches** on the same page. Deleting removes all its sessions and allocations and cannot be undone, so take a backup first. The rotation history (which halls each person has already had) is kept, so later allocations still count the duties that were done; use Settings → Restart Rotation if you want to clear it. The Allocation Batches list itself has no delete button, so a batch cannot be removed by a stray click.

6. Allocating invigilators to a session

Open a batch to reach its **Allocation Workspace**. Each session appears as a tab (for example *02 Nov 2026 FN*).

1. Select a session tab and click **Select Staff & Generate**.
2. Choose the staff who are on duty and the halls in use for this session. **The number of staff must equal the number of halls**; HIAS tells you if they differ. You can:
   - tick people one by one, or use **Select all** for a department;
   - type part of a **name or Staff ID** in the search box to find someone quickly, and part of a **hall code or block** in the halls search box to find halls the same way;
   - click **Upload Excel** to select everyone on a duty list at once (see below);
   - click **Copy previous session**, or pick a session under **Copy from a session…**, to start from the staff and halls ticked in an earlier session (see below).
3. Click **Generate Allocation**. HIAS assigns everyone at once: each person gets a hall they have not yet had in their current round of halls, drawn at random rather than in a fixed order, and halls are spread evenly over time. Generating the same session again with the same staff and halls gives the same result.
4. Review the list. To change an assignment, click the pencil on that row and choose a hall: the invigilator of that hall takes this person's hall, so the two **swap**. HIAS only offers swaps that keep the rules satisfied for **both** people; the other halls are marked *Current*, *In cycle* or *Swap blocked* (point at one to see why). **Regenerate** starts the session over; to start again from the same staff and halls, choose **This session (current draft)** under **Copy from a session…**.
5. Click **Confirm Allocation**. HIAS runs all rule checks; if something is wrong it explains what, and nothing is saved until it is fixed.
6. Click **Export PDF** to save the printable allocation sheet for this session. It is an A4 portrait page with two blank columns, **Adjusted By** (to write the name of the staff member who covers if the allotted person cannot come) and **Staff Signature**.



A confirmed session. Rows you changed by hand are marked "Admin Edited" instead of "Auto-generated" (in a swap, both people's rows); swapping back to the generated halls removes the mark.

Uploading a duty list from Excel

Instead of ticking people one by one, upload an Excel (`.xlsx` , `.xls`) or `.csv` file listing the invigilators on duty:

1. In the selection window, click **Download template**. It is an Excel sheet of all active staff with the columns **Staff ID**, **Name**, **Department** and **Hall**.
2. Delete the rows of people who are not on duty. Optionally type a hall code (for example `H101`) in the Hall column to choose which halls are open. Save the file.
3. Click **Upload Excel** and choose the file. Everyone listed is selected (and the halls, if a Hall column is filled in); anything already ticked is replaced.
4. Check the summary. Any row that could not be used is listed with the reason, for example an unknown Staff ID, a name shared by two people, or a person listed twice.

Your own duty list works too, as long as its first row has a **Staff ID** column (or a **Name** column). A sheet with no headings is read as a list of Staff IDs in the first column. The Hall column only decides **which halls are used**; HIAS still decides who goes to which hall, so the rotation stays fair.

Copying the staff and halls of another session

When the people on duty barely change from one session to the next, start from an earlier session's ticks instead of ticking everyone again:

- **Copy previous session** ticks the same staff and halls as the nearest earlier session of this batch (for the first session of a batch, the latest earlier session of any batch).
- **Copy from a session…** lists every session that has an allocation, this batch first and newest first. After **Regenerate** it also offers **This session (current draft)**.

The copy replaces what is ticked; then add or remove people as usual. Staff and halls that are no longer active are left out, and the summary says how many. Only the **selection** is copied, never who had which hall: **Generate Allocation** draws everyone's hall afresh, exactly as if you had ticked them by hand.

> **Why confirming matters.** The rotation only moves forward when a session is **confirmed**, and HIAS confirms sessions strictly in order: a session can be confirmed only when every earlier session of the batch is confirmed. Sessions you add or reschedule later take their place in date order among the sessions not yet confirmed; confirmed sessions never move.

The rules HIAS checks

Rule	Meaning
R1	A staff member does not return to a hall until they have been through the other halls in the rotation. When the people on duty change a lot, this is occasionally impossible for everyone; HIAS then keeps repeats to the minimum and shows them as yellow **"Unavoidable repeat"** warnings, which do not block confirmation.
R2	A hall gets only one invigilator per session.
R3	A staff member is in only one hall per session.
R4	Every earlier session of the batch must be confirmed first.
R5 / R6	Inactive staff and inactive halls are never used.
R7	The number of staff equals the number of halls.

7. Student seating

Students sit in the halls of the session's **invigilator allocation**, so allocate the invigilators first (section 6). Then, in the Allocation Workspace, pick the session tab and switch from **Invigilators** to **Student Seating**.

1. Click **Select Students & Configure Seating**. Tick the students who write an exam in this session: search by roll number, register number or name, tick a **whole batch** at once, click **Upload list** to tick everyone on an Excel or CSV list of roll or register numbers, or use **Copy from session…** to start from another session's students. **Next: Configure Halls** stays disabled while more students are ticked than the halls have seats.
2. In **Seating Configuration**, choose the **Allocation method** (below). HIAS sets up the halls automatically; change any hall if you need to. The preview on the right updates as you go: the seated and unallocated counts, a table of every hall with the classes in it, and the bench grid of the hall you click.

3. Set **Classes per hall**: the **Target** number of different classes (papers) a hall may hold, and **up to** how many if the target would leave students without a seat. The default is 3, up to 4.
4. Click **Confirm & Save**. If some students cannot be seated, HIAS lists them with the reason and asks before saving; the list is kept with the plan.

Allocation methods

Method	How students are seated
Roll number order	Boys and girls together. Each seat column follows roll numbers whatever the gender: `102349` (a boy) is followed by `102350` (a girl). Each hall is just **Use** or **Not used**. A boy and a girl of different classes may share a bench.
Separate halls	Each hall holds only boys or only girls. HIAS tries every combination of halls to find the split that seats the most students with the fewest empty seats.
Same pattern in every hall	One **bench pattern** repeated bench by bench in every hall used, for example *Boy + Boy, Girl + Girl*: bench 1 boys, bench 2 girls, bench 3 boys, and so on.
Custom per hall	Each hall on its own: **Boys**, **Girls**, **Boys & Girls** (with its own bench pattern) or **Not used**.

A bench pattern is made of **Boy + Boy** and **Girl + Girl** benches only: except in Roll number order, a boy and a girl never share a bench. Click **+ Bench** to add a bench to the pattern, the cross to remove one, or the drop-down to change it. **Auto-assign halls** puts the halls back to HIAS's own choice for the method.

How the seats are filled

Each column of benches has a left-seat column and a right-seat column. HIAS fills each seat column front to back with **one class**, in roll-number order, and the classes take the seat columns in turn (the largest class first), each continuing from where it stopped; the next hall carries on with the next class. So the two students on a bench are always of different classes: **two students of the same class never share a bench**. If a class runs out part-way down a column, the next class in turn continues it. Only when that would leave a student without a seat does HIAS move students between seats to fit them in.

> **Unallocated students.** A student is left unallocated only when there is no seat for them: not enough seats, no free seat beside a student of another class, or (in the methods that keep boys and girls apart) no seat for their gender. Each is listed with the reason. To seat them, add halls to the session, allow more classes per hall, or change the hall settings.

Printing

Once a plan is saved, the Student Seating tab shows each hall's bench grid and these PDFs:

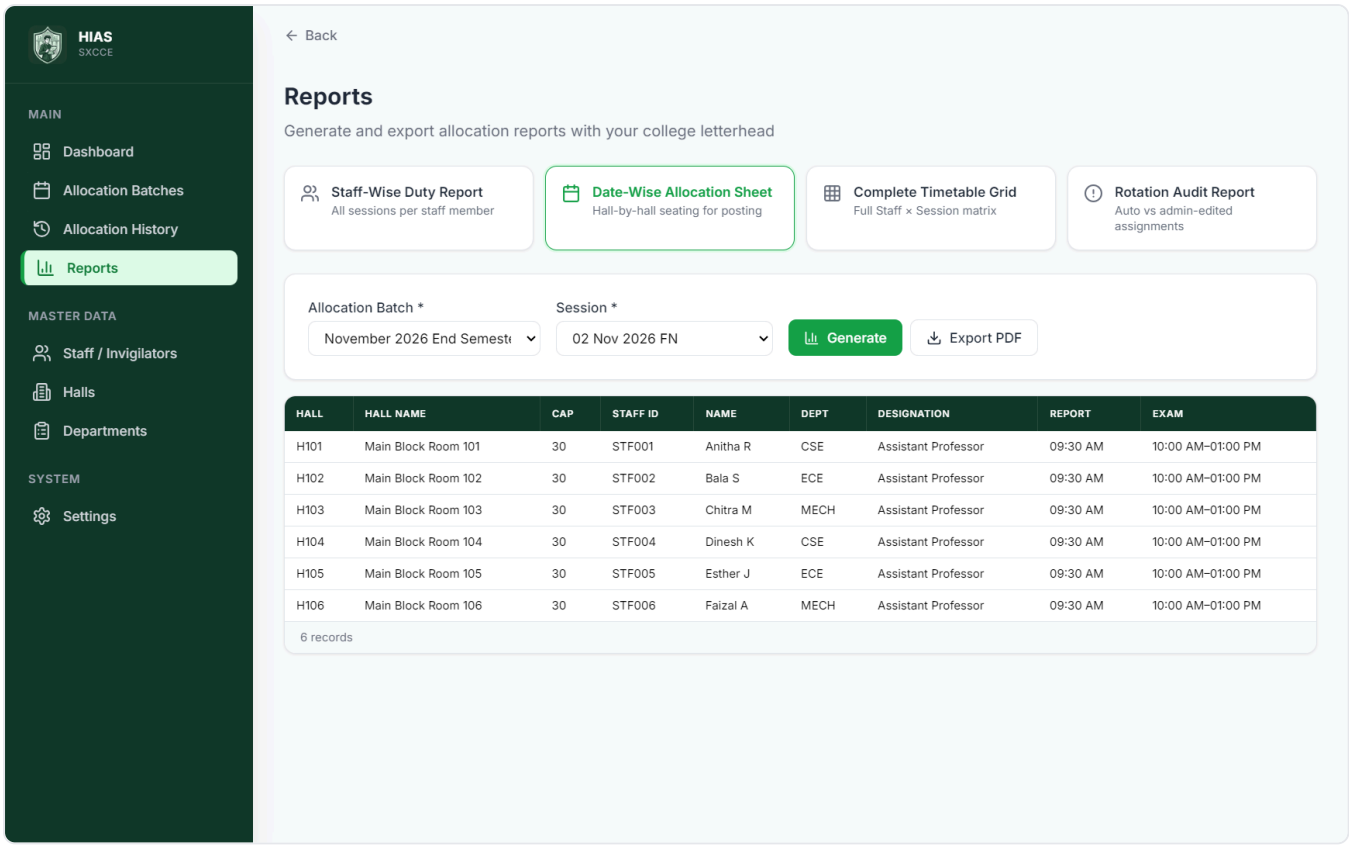
Button	What it prints
Seating Plan	One landscape page per hall laid out like the screen: *Column 1, Column 2…* across, *R1, R2…* down, the board at the top, and a card for each bench with its number and the roll numbers on its left and right seats. It is made for **black-and-white printers**: roll numbers are in bold black, and the years are told apart by the box (white with a solid border, grey, or white with a dashed border), explained in the key at the bottom. In a hall of boys and girls, each seat carries a **B** or **G**.
Attendance Sheets	One page per hall: the students in bench order with a column for their signature, and spaces for the present and absent counts and the invigilator's signature.
Student List	Every student by batch and roll number with their hall, bench and seat, for the notice board.
Unallocated → Print / PDF	The students left without a seat, with the reason.

Edit Configuration reopens the saved settings, **Reselect Students** changes who is seated, and **Clear** removes the session's seating plan.

8. Reports

Open **Reports**, choose a report type, choose the filters, click **Generate** to preview it, then **Export PDF**. Each PDF carries the college name and the time it was generated. When you export, Windows asks where to save the file (the Downloads folder is suggested).

Report	Use it for	Filters
Staff-Wise Duty Report	Every duty of each staff member: date, session, hall and timings.	Optional: one staff member
Date-Wise Allocation Sheet	The hall-by-hall list for one session, for the notice board. The title names the date and session.	Batch and session
Complete Timetable Grid	One page showing every staff member against every session of a batch.	Batch
Rotation Audit Report	Which assignments were generated automatically and which were changed by an admin (swaps).	Batch



Date-Wise Allocation Sheet preview, ready to export.

9. Allocation history

Allocation History lets you look up any past confirmed allocation. Filter by batch, session, staff member or hall and click **Search Records**. This is the quickest way to answer "Where was this person on that day?".

10. Backup and restore

> **HIAS keeps its data only on this computer.** If the computer fails and there is no backup, the data cannot be recovered. Take a backup after every allocation batch and keep it on a pen drive or another computer.

To take a backup

1. Open **Settings** → **Backup & Restore** and click **Backup Database**.
2. Choose where to save the file (for example a pen drive). The file is named `hias-backup.db` by default; adding the date to the name helps.

To restore a backup

1. Open **Settings** → **Backup & Restore** and click **Restore Database**.
2. Choose the backup file and confirm. HIAS first checks that the file really is a HIAS backup (if not, it says so and changes nothing), saves a safety copy of the current data in the data folder (named like

`eias.db.before-restore-20261101-093000.bak`, with the date and time), and then replaces it.

3. HIAS restarts by itself a moment later and opens with the restored data.

The same steps move HIAS to a new computer: back up on the old computer, install HIAS on the new one, and restore. Everything (staff, students, halls, batches, allocations, seating plans and the rotation) travels in that one file. To go back to the data from before a restore, restore the safety copy the same way.

11. Restarting the rotation

Settings → **Restart Rotation** clears the whole rotation history, so every staff member starts a fresh round with every hall open to them again. Use it only at a genuine new start, such as the beginning of an academic year. You must type `RESTART` to confirm. Past allocations and reports are not deleted; only the rotation memory is cleared.

12. Troubleshooting and FAQ

Problem	What to do
"Staff count must equal hall count"	Select the same number of staff and halls for the session.
A staff member or hall is missing from the selection list	Check it is marked **Active** in Staff or Halls.
Confirm is refused with a rule message (R1–R7)	Read the message; it names the person or hall. Edit that row, or regenerate, then confirm again.
"No bench layout" for a hall in Student Seating	Open **Halls**, edit the hall and set its Columns and Rows under Bench layout, or mark the hall Not used for seating.
Next: Configure Halls is greyed out	More students are ticked than this session's halls have seats. Untick some, or add halls to the session's invigilator allocation.
Students are left unallocated although there are empty seats	Two students of the same class never share a bench and a hall holds only a few classes. Allow more **Classes per hall**, add a hall, or try **Roll number order**.
Windows says "Windows protected your PC" when installing	Click **More info** → **Run anyway**. The installer is not code-signed.
The app shows an error box at start-up	Note the message, then restore the latest backup (section 10). The message is shown because a silent failure would give no clue.
Can two computers share the same data?	No. Each installation has its own data. Use one computer for allocation, and move data with backup and restore if needed.
Does HIAS need the internet?	No. It works completely offline: Wi-Fi and network cables can be disconnected. Reports and PDFs are made on this computer.
Is there a login or password?	No. HIAS is meant for one trusted computer; whoever can use its Windows account can use HIAS. Lock the computer (Windows key + L) when you leave it.
The data file was deleted or the computer was replaced	HIAS starts with a new, empty database. Restore your latest backup (section 10) to get everything back. Without a backup the data cannot be recovered, which is why regular backups matter.